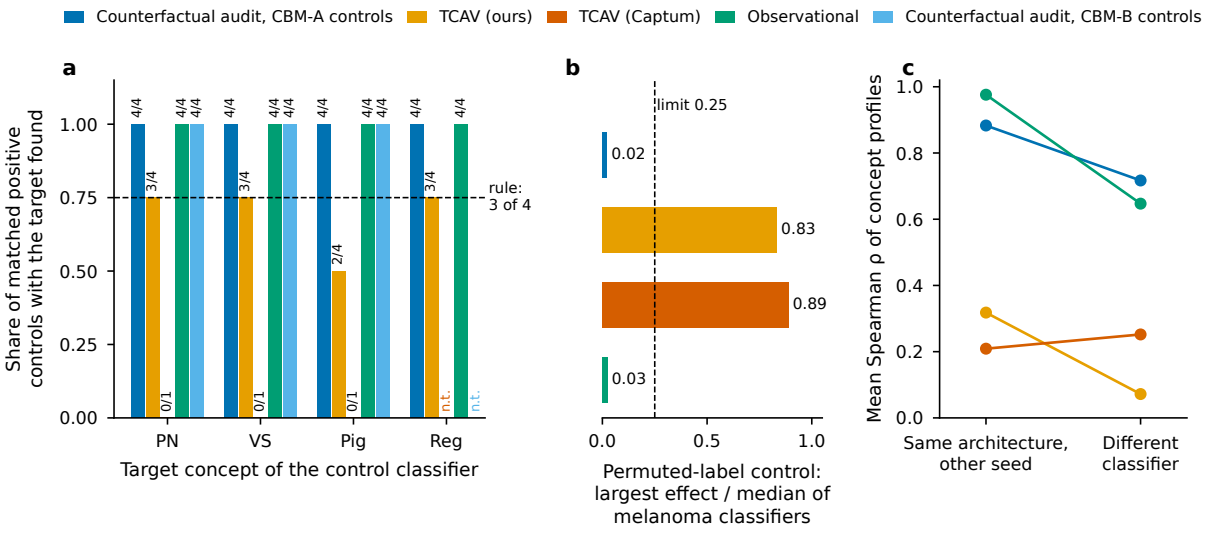
Counterfactual audit, CBM-A controls
TCAV (ours)
TCAV (Captum)
Observational
Counterfactual audit, CBM-B controls
a
Share of matched positive controls with the target found
1.00
0.75
0.50
0.25
0.00
rule: 3 of 4
4/4
3/4
0/1
2/4
n.t.
PN
VS
Pig
Reg
Target concept of the control classifier
b
limit 0.25
0.02
0.83
0.89
0.03
0.0
0.5
1.0
Permuted-label control: largest effect / median of melanoma classifiers
c
Mean Spearman ρ of concept profiles
1.0
0.8
0.6
0.4
0.2
0.0
Same architecture, other seed
Different classifier